

**SIMATS ENGINEERING**  
**ASSESSMENT TOOL 2: Concept Mapping**

**COURSE NAME:**

Cloud Computing and  
Big Data Analytics

**COURSE CODE:**

CSA15

**COURSE OUTCOME COVERED**

**CO1:** Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)  
Question Count: 4

**Duration: 60 Minutes**

**Total Marks: 20**

**Code of Conduct**

I N. Jaffari / P2521200 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: N. Jaffari

| Criteria  | Concept Identification (1.25) | Relationship Mapping (1.25) | Hierarchy and Organization (1) | Technical Relevance (1) | Visual Clarity (0.5) | TOTAL (5) |
|-----------|-------------------------------|-----------------------------|--------------------------------|-------------------------|----------------------|-----------|
| Weightage | 25%                           | 25%                         | 20%                            | 20%                     | 10%                  | 100%      |
| Q1        |                               |                             |                                |                         |                      |           |
| Q2        |                               |                             |                                |                         |                      |           |
| Q3        |                               |                             |                                |                         |                      |           |
| Q4        |                               |                             |                                |                         |                      |           |

Signature of the Faculty

**Total Marks Awarded:**

## Assessment Tasks

### Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Architecture: Include front-end, back-end, client devices, data centers, and distributed servers.
2. Concept Map 2: Characteristics of Cloud Computing: Connect on-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service.
3. Concept Map 3: Deployment Models: Map public cloud, private cloud, hybrid cloud, and community cloud with usage scenarios.
4. Concept Map 4: Cloud Benefits vs Challenges: Show advantages (cost efficiency, scalability, flexibility) and limitations (security, downtime, vendor lock-in).

### Expected concept nodes:

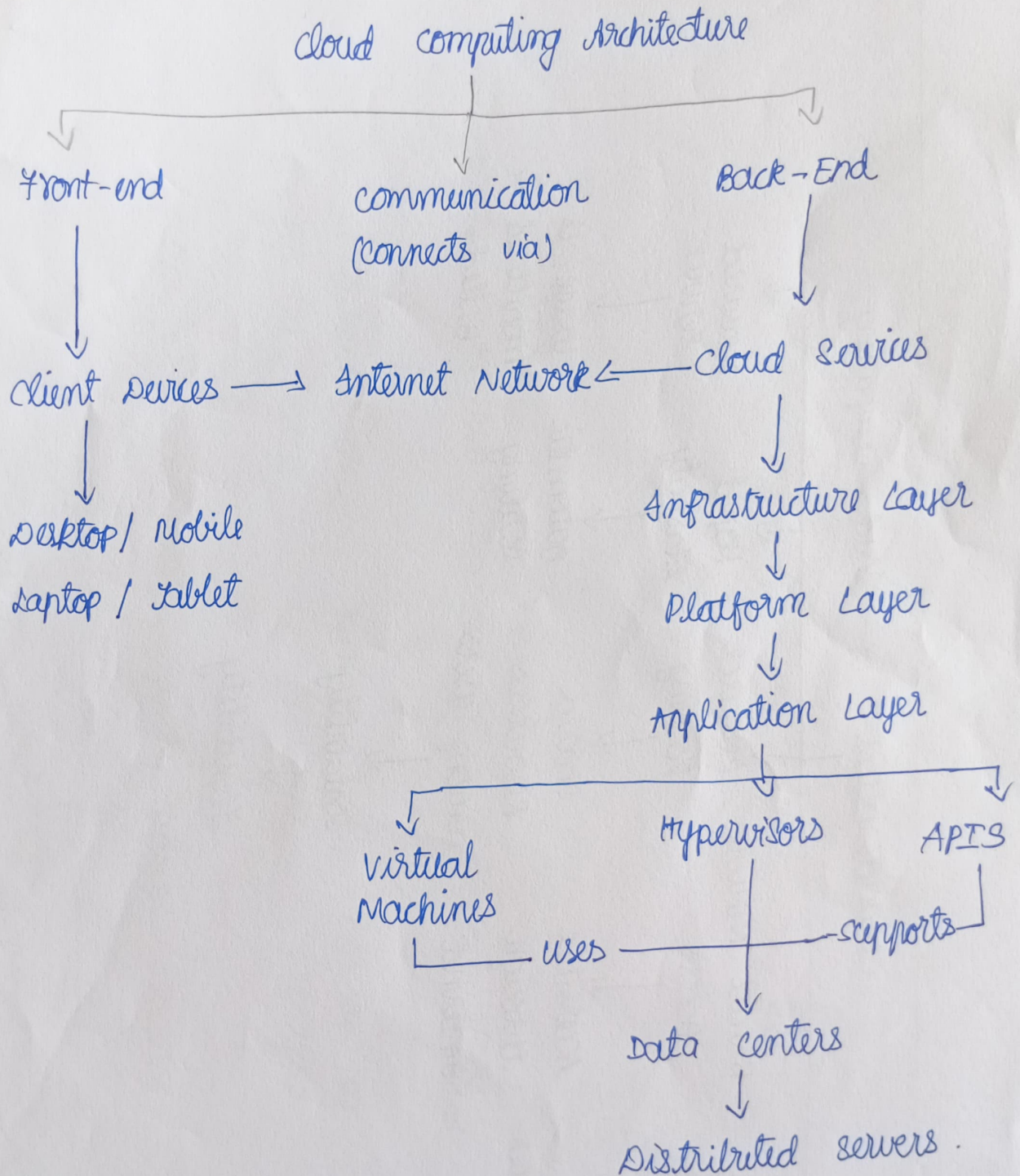
- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

### Concept Mapping Rubric

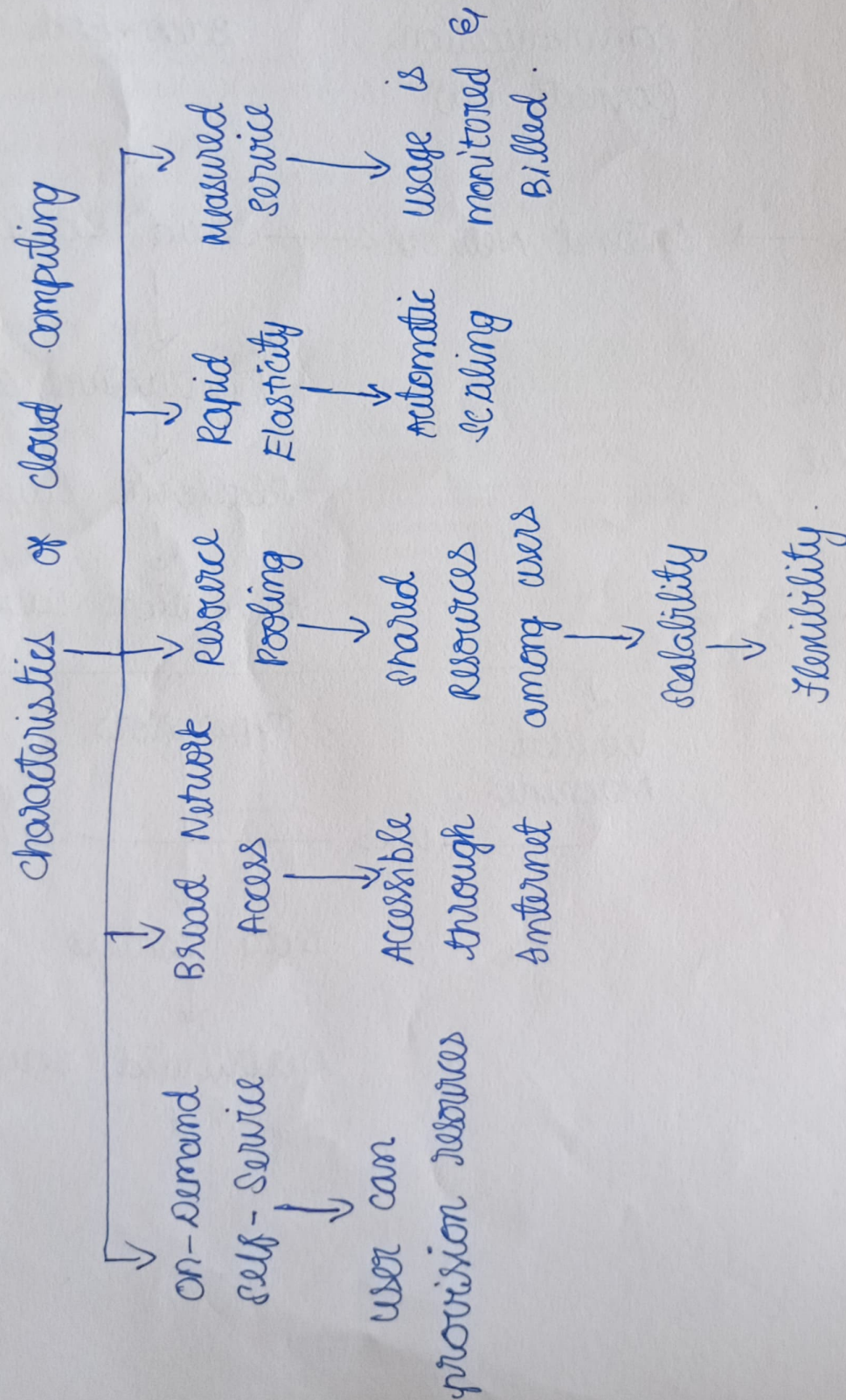
| Criteria                   | Weightage | Level 4 - Exemplary                                | Level 3 - Proficient                   | Level 2 - Developing                | Level 1 - Beginning                |
|----------------------------|-----------|----------------------------------------------------|----------------------------------------|-------------------------------------|------------------------------------|
| Concept Identification     | 25%       | All major concepts are accurately identified       | Most concepts are identified correctly | Limited concepts are identified     | Essential concepts are missing     |
| Relationship Mapping       | 25%       | Links between concepts are accurate and meaningful | Most links are correct                 | Some links are weak or unclear      | Relationships are mostly incorrect |
| Hierarchy and Organization | 20%       | Excellent structure with clear hierarchy           | Mostly organized with minor issues     | Basic structure with weak hierarchy | No logical organization            |
| Technical Relevance        | 20%       | Map strongly reflects cloud course content         | Mostly aligned to topic                | Partially aligned to topic          | Weak alignment to topic            |
| Visual Clarity             | 10%       | Neat, readable, and easy to interpret              | Generally readable                     | Somewhat cluttered                  | Poorly presented                   |

# Concept mapping tasks.

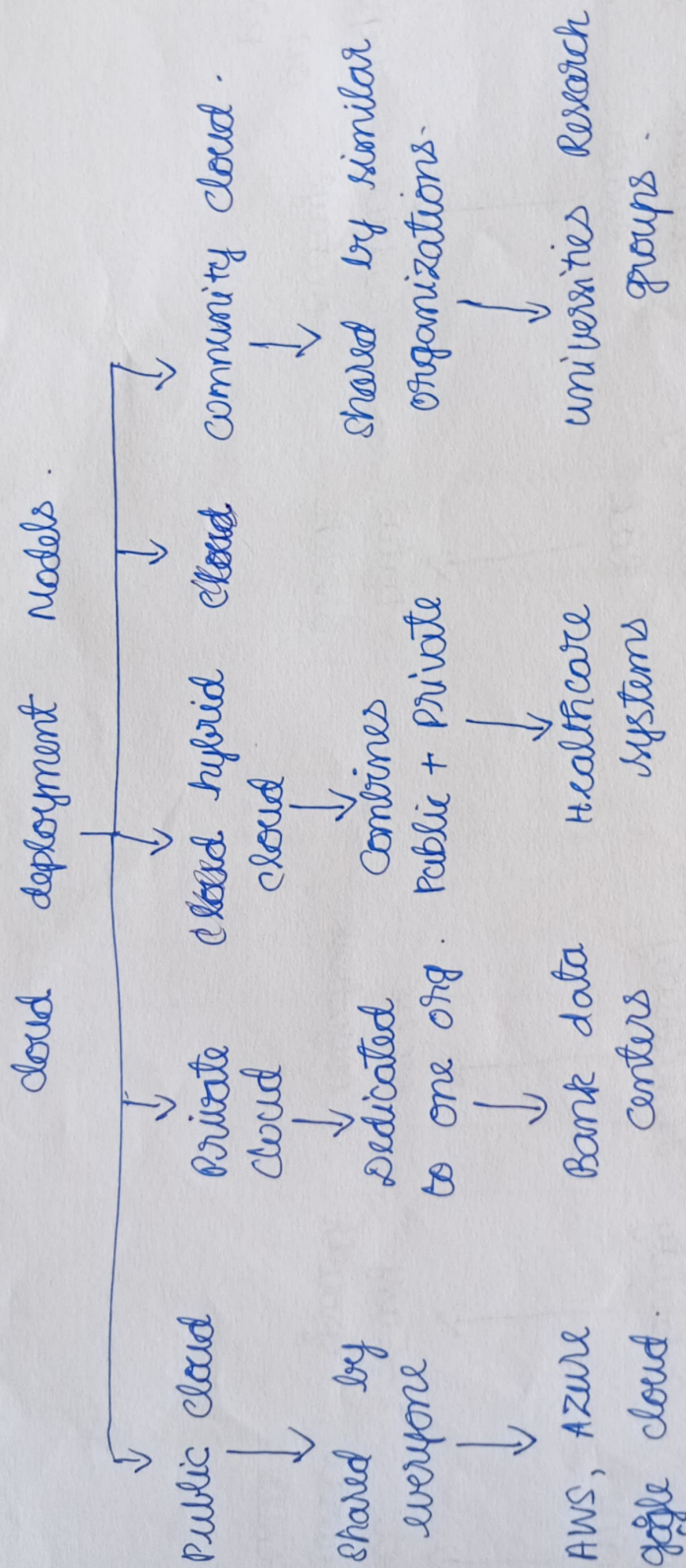
## 1) Cloud Computing Architecture.



## 2) Characteristics of cloud computing



### 3) cloud Deployment Models.



#### 4) cloud Benefits vs challenges.

